AgriHGT: Heterogeneous Graph Transformer for Cross-Crop Leaf Disease Relation Modeling in Tomato, Papaya, Zucchini, and Bottle Gourd

***Abstract*—Leaf-disease classification across multiple crops is challenging because disease symptoms may appear visually similar, but relationships among crops, diseases, and disease groups are usually not considered by image-only classifiers. AgriHGT, a heterogeneous graph transformer framework, was developed to combine visual representations with structured crop-disease information. The Agri-Vision4 dataset was used, covering Tomato, Papaya, Zucchini, and Bottle Gourd with 28 crop-specific classes. After duplicate removal, 26,866 images were retained and divided using a duplicate-aware and support-controlled splitting procedure. Seven pretrained visual backbones were evaluated, including supervised and self-supervised models. Extracted image features were represented as leaf-image nodes and connected with crop, disease, and disease-category nodes through typed relations. HGT message passing was combined with a residual visual branch and a learned fusion mechanism for final disease prediction. The strongest observed result was obtained using DINOv3 ViT-B/16 with two HGT layers, achieving 93.87% test accuracy and 85.88% macro-F1. The corresponding feature-only reference achieved 91.63% accuracy and 80.27% macro-F1, showing that relational modelling provided additional benefit beyond the visual representation alone. Depth and component ablation experiments further showed the importance of graph message passing, the residual visual path, and learned feature fusion. The results indicate that combining self-supervised visual features with heterogeneous graph information is a promising approach for multi-crop leaf-disease classification,although further evaluation on completely unseen leaf images and across multiple random seeds is required for broader generalization.**

***Index Terms*—Plant disease classification, heterogeneous graph transformer, HGT, DINOv3, self-supervised learning, graph neural network, multi-crop classification.**

I. Introduction

Plant disease classification from leaf images is an important computer-vision problem because plant diseases can reduce crop yield, damage crop quality, and create economic loss for farmers. Early and correct identification of a disease is therefore important for timely treatment and better crop management. However, manual disease diagnosis usually depends on farmers, agricultural officers, or plant pathology experts. This process can be difficult when expert support is not available, when several diseases produce similar visual symptoms, or when a large number of plants must be checked. Automated plant disease classification can therefore provide useful decision support by identifying diseases directly from leaf images.

Plant disease symptoms are commonly visible through changes in leaf colour, texture, shape, spots, holes, lesions, or other damaged regions. These visual patterns have made leaf images suitable for deep-learning based disease recognition. Convolutional Neural Networks (CNNs) have been widely used for this task because useful image features can be learned automatically instead of being manually designed. CNN-based methods have produced strong results on several plant disease datasets [1], [2]. Transfer learning has also been widely applied, where a model pretrained on a large image dataset is adapted to plant disease images. Architectures such as ResNet, DenseNet, EfficientNet, and MobileNet have been successfully used in this way [3], [4].

More recently, Vision Transformers (ViTs) have been studied for plant disease recognition. Unlike a conventional CNN, a ViT divides an image into patches and uses self-attention to learn relationships among those patches. This allows information from distant parts of an image to be considered together. ViT-based and hybrid CNN–transformer models have shown strong performance on plant-disease datasets [5], [6]. Swin Transformer and other hierarchical transformer architectures have also been used to capture local and global visual information. Self-supervised methods have further been explored to reduce dependence on manually labelled data and to learn stronger image representations before supervised fine-tuning [7], [8].

Another important research direction is multi-crop disease classification. Instead of developing a separate model for every crop, several crop species and their diseases can be included in one classification system. Recent multi-crop models have classified diseases from crops such as rice, wheat, bean, tea, banana, sugarcane, tomato, potato, and other plants [9]–[11]. This approach is useful because a single model can cover more crop and disease classes. However, most multi-crop methods still treat the problem as standard image classification. Each leaf image is passed through a CNN or transformer, and one class is predicted. The relationships among the crop, its possible diseases, visually similar leaves, and related disease groups are usually not represented explicitly.

This limitation is important because plant diseases are not completely independent concepts. A disease occurs on a particular host crop, and different diseases may belong to similar biological or symptom-based groups such as fungal, bacterial, viral, pest-related, or nutrient deficiency conditions. Similar disease patterns may also appear in different crops. Therefore, useful information may exist not only inside an individual image but also in the relationships among images, crops, diseases, and disease categories. A conventional image classifier does not directly represent these relationships.

Graph Neural Networks (GNNs) provide a way to learn from such relationships. In a graph, information is represented using nodes and edges, and connected nodes can exchange information through message passing. GNNs have

already been applied to plant-disease problems. For example, graph-based methods have been used to combine regional image features, model visually related leaves, or improve local feature aggregation [12], [13]. Agricultural knowledge graphs have also been developed to represent crops, diseases, pests, symptoms, and other agricultural entities [14]. Other studies have combined visual features with knowledge-graph information for plant disease classification [15].

However, these approaches solve only part of the problem. Image-based GNN methods commonly construct graphs between image regions or between similar leaf images. Agricultural knowledge graphs usually focus on knowledge storage, retrieval, question answering, or semantic reasoning. Knowledge-graph fusion methods may add graph embeddings to visual features, but individual leaf images are not always represented as one node type together with explicit crop, disease, and disease-category nodes in a common message-passing graph. Therefore, multi-crop image classification and agricultural relational knowledge are still often handled as separate problems.

Another important issue is evaluation of the graph contribution. A complex graph model may produce better accuracy simply because it contains more parameters or additional components. Improvement cannot automatically be attributed to graph message passing. A controlled comparison is therefore needed between a feature-only model, a model without message passing, and models with different graph depths. Ablation experiments are also required to determine whether image similarity, crop information, disease relationships, taxonomy information, residual visual features, and learned fusion actually contribute to classification performance.

To address these limitations, a two-stage framework called *AgriHGT* is proposed. In the first stage, a pretrained visual backbone is adapted using the training data and is then used to extract fixed visual embeddings from the leaf images. Seven backbones are evaluated: ResNet50, ConvNeXt-Tiny, EfficientNet-B3, Swin-Tiny, DenseNet121, DINOv2 ViT-B/14, and DINOv3 ViT-B/16. These backbones cover supervised CNNs, transformer-based architectures, and self-supervised vision-transformer representations.

In the second stage, the extracted image embeddings are placed inside a heterogeneous graph. Four different node types are represented: `leaf_image`, `crop`, `disease`, and`disease_category`. Different edge types describe different relationships, including crop membership, crop–disease relationships, disease-category membership, image similarity, and selected image–disease connections. A Heterogeneous Graph Transformer (HGT) [16] is then used so that different node types and relation types can be processed separately during message passing.

The main objectives and contributions of this study are as follows:

1) To construct a heterogeneous graph using leaf images, crops, diseases, and disease categories so that visual and relational information can be processed together.
2) To evaluate seven visual backbones including ResNet50, ConvNeXt-Tiny, EfficientNet-B3, Swin-Tiny, DenseNet121, DINOv2 ViT-B/14, and DINOv3 ViT-B/16.
3) To compare zero, one, and two layer HGT configurations to examine whether graph message passing provides additional value beyond the original backbone features.
4) To compare AgriHGT a feature-only logistic-regression baseline to measure the practical benefit of the complete relational framework.
5) To perform ablation experiments are performed to study the importance of image-similarity edges, crop information, disease taxonomy, image–disease grounding, learnable meta-node representations, residual visual features, and the learned fusion gate.
6) To compare cross-crop and crop-specific graph settings to examine whether shared multi-crop relational structure provides the same benefit for every crop.

The remainder of this paper is organized as follows. Section II reviews recent CNN, transformer, self-supervised, multi-crop, GNN, and knowledge-graph approaches. The methodology section describes the dataset preparation, backbone adaptation, graph construction, HGT architecture, training procedure, and ablation settings. The experimental-results section reports backbone comparisons, message-passing-depth results, class-level performance, graph-component ablations, and cross-crop experiments. Finally, the conclusion summarizes the main results and identifies directions for future work.

II. Related Work

A. CNN and Supervised Transfer-Learning Approaches

A transfer-learning comparison for field and greenhouse rust detection was conducted by Shahoveisi et al. [2]. A dataset of 1,764 images containing sunflower, dry bean, and field-pea leaves was used, with 907 healthy and 857 rust-infected samples captured under variable backgrounds and lighting conditions. Xception, ResNet50, EfficientNetB4, and MobileNetV2 were evaluated under different optimizers and learning rates, followed by ten-fold cross-validation for the best configuration. EfficientNetB4 produced an average accuracy of 94.29%, while the selected configuration achieved $94.34\% \pm 0.010$ under cross-validation. Grad-CAM was used to confirm attention to rust pustules rather than irrelevant background regions. The work addressed robustness beyond controlled laboratory imagery and included several crop hosts, but only binary rust detection was considered and semantic relationships among crops and diseases were not modelled.

A deep transfer-learning ensemble was developed by Shafik et al. [3] for classification of 15 selected PlantVillage disease classes. Nine pretrained architectures, including DenseNet201, ResNet101, ResNet50, EfficientNetB7, NASNetMobile, and ConvNeXtSmall, were used for deep feature extraction, followed by logistic-regression-based early fusion and lead-voting ensembles. PlantVillage contains 54,305 images across 38 original categories, but 15 classes were selected for the reported experiments. PDDNet-AE and PDDNet-LVE achieved accuracies of 96.74% and 97.79%, respectively. A dedicated post-hoc XAI analysis was not reported. Improved ensemble robustness was demonstrated, although reliance on controlled

PlantVillage imagery, single-leaf inputs, and several pretrained models increased the gap between benchmark performance and lightweight field deployment.

A fine-tuned EfficientNet-B0 model was evaluated by Ali et al. [4] using the PlantVillage dataset and an Apple PlantVillage subset containing healthy leaves and three apple diseases. Stratified splitting, class weighting, full-network fine-tuning, dropout, regularization, and augmentation through geometric and photometric transformations were used to reduce imbalance and overfitting. Test accuracies of 99.69% on the apple subset and 99.78% on PlantVillage were obtained, while relatively low memory and FLOP requirements were retained. A dedicated Grad-CAM or comparable XAI analysis was not included. The approach effectively addressed fine-grained apple-disease classification and computational efficiency, but performance remained dominated by controlled benchmark imagery and no relational information between crop and disease classes was used.

Cross-dataset generalization was examined by Mahapatra et al. [17] through an attention-based CNN. Digipathos, Northern Leaf Blight, PlantVillage, PlantDoc, and CD&S datasets were used for corn and potato disease classification. Both intra-dataset and cross-dataset experiments were performed, and background-removed CD&S images were additionally used to study domain shift caused by image backgrounds. The highest intra-dataset accuracy was 99.38% for potato PlantVillage images, whereas the strongest average cross-dataset accuracy was 82.93% for corn. A dedicated post-hoc XAI evaluation was not reported. The work directly addressed dataset-domain generalization, but the substantial reduction from intra-dataset to cross-dataset performance demonstrated that visual models remain sensitive to acquisition conditions, and only two crop domains were examined.

B. Vision Transformer and CNN–Transformer Approaches

A lightweight Vision Transformer framework was investigated by Borhani et al. [5] using Wheat Rust Classification, Rice Leaf Disease, and PlantVillage datasets. CNN, ViT, and combined CNN–ViT architectures were trained from scratch at several image resolutions to compare accuracy, computational complexity, and prediction speed. The ViT-based configuration generally produced the strongest classification performance, while attention layers improved recognition at the cost of slower prediction. No dedicated post-hoc explainability method was reported. The work demonstrated that transformer attention can be useful for datasets ranging from small to large scale, but interactions remained restricted to visual patches within individual images and no relations among crop or disease entities were represented.

A lightweight CNN–transformer architecture was proposed by Thakur et al. [6] and evaluated across five public plant-disease datasets. Initial VGG16 convolutional blocks and an Inception module were combined with four transformer encoder modules, producing approximately 0.85 million trainable parameters. Accuracy of 98.86% was reported on PlantVillage, while 89.24% was obtained on the more challenging Embrapa dataset. Grad-CAM and LIME were used to examine disease-relevant image regions and provide complementary explanations. The framework addressed the computational cost of large transformers while maintaining strong classification accuracy; however, multiple crop classes were still treated as independent image-label pairs rather than semantically connected entities.

Joint disease localization and classification were addressed by Hemalatha and Jayachandran [18] using a multitask Plant Disease Localization and Classification Vision Transformer (PDLC-ViT). PlantVillage was used for classification, whereas PlantDoc and corresponding masks were used for disease localization. Affine augmentation, grid-search hyperparameter tuning, early stopping, co-scale representations, co-attention, and cross-attention were incorporated into the framework. Accuracy of 99.97%, mean average precision of 99.18%, and mean average recall of 99.11% were reported. Grad-CAM-based gradient activation maps were used for explainability. Simultaneous localization and classification were successfully addressed, but separate datasets were used for the two tasks, introducing possible domain differences, and no explicit relationships among crops or diseases were incorporated.

A hybrid CNN–ViT framework was developed by Aboelenin et al. [19] using public apple and corn datasets, each containing four classes. Images were resized to 128×128, augmentation was applied, and features extracted by VGG16, Inception-v3, and DenseNet201 were combined with ViT-based self-attention and an MLP classifier. Accuracies of 99.24% and 98.00% were achieved for apple and corn, respectively. A dedicated post-hoc XAI analysis such as Grad-CAM was not included in the reported evaluation. Local and global visual feature extraction was effectively combined, but computational complexity increased because several backbones were used and only two crop-specific datasets were evaluated without explicit cross-crop relational modelling.

C. Self-Supervised, Zero-Shot, and Cross-Domain Recognition

Reduced dependence on labelled data was investigated by Wang et al. [7] through masked-autoencoder-based self-supervised learning. The CCMT field dataset and an independently collected potato dataset were used; CCMT contains tens of thousands of images from multiple crops and disease categories. Rotation, random cropping, and horizontal flipping were used for augmentation. A masked autoencoder was combined with a ViT backbone, GRU, and CBAM attention, producing approximately 95.4% accuracy on CCMT and approximately 99.6% on the collected dataset. CBAM provided internal attention weighting, but a dedicated post-hoc XAI evaluation was not reported. The method reduced label dependence and improved representation learning, although final classification remained image based and no explicit crop–disease relational structure was learned.

A self-supervised contrastive-learning framework with domain adaptation was proposed by Zhao et al. [8]. PlantVillage, PlantDoc, and an augmented leaf-disease dataset were used to create a setting with large quantities of distribution-shifted unlabelled data and limited labelled target data. Contrastive pretraining was followed by a domain-adaptation layer and

supervised fine-tuning, allowing visual representations from the two domains to be aligned. A highest reported accuracy of 90.52% was obtained, with additional experiments showing sensitivity to dataset volume and source–target mixing ratios. A dedicated image-level XAI analysis was not reported. The method directly addressed scarcity of labelled data and domain shift, but relational knowledge among crop species, disease concepts, and individual images was not represented.

Zero-shot cross-crop disease recognition was investigated by Kumar et al. [20] using a PlantVillage subset containing tomato, potato, and pepper classes. Thirteen manually annotated semantic attributes describing colour, shape, texture, spots, discoloration, and other symptoms were concatenated with pretrained CNN or ViT visual features. In a three-class tomato-to-potato experiment, ResNet18 reached 99.95% accuracy with semantic attributes compared with 62.73% without the attributes; a more difficult four-class setting produced substantially lower performance, with ViT-B/16 reaching 69.45%. Semantic attributes and feature visualizations increased interpretability, although no conventional Grad-CAM evaluation formed the main explanation mechanism. The approach demonstrated the value of explicit disease semantics for cross-crop recognition, but manual attribute annotation limits scalability and semantic information was concatenated rather than propagated through a relational graph.

Cross-crop generalization to completely unseen crop domains was examined by Wu et al. [21] through Text-guided feature Disentanglement Contrast (TDC). PlantVillage was reorganized into six shared disease concepts across eight crop domains, producing 21,521 original images and 19,369 background-removed training images. A dual-branch ViT-B/16 encoder, frozen CLIP text encoder, contrastive disentanglement, domain-adversarial learning, semantic text anchors, and prototype-based classification were used to separate disease information from crop-specific appearance. Seen-domain accuracy of 98.04%, unseen-domain accuracy of 74.29%, and a harmonic mean of 84.53% were reported; external strawberry powdery-mildew evaluation reached 81.99%. Semantic and prompt analyses supported interpretability, although conventional Grad-CAM was not the central XAI method. Cross-crop generalization was directly addressed, but only disease concepts shared across crop domains could be evaluated and no explicit image–crop–disease graph was constructed.

D. Multi-Crop Disease Classification

A unified lightweight multi-crop classifier named Plantention was proposed by Subburaj et al. [9]. Five public datasets representing rice, wheat, tea, banana, and sugarcane were used rather than training a separate architecture for every crop. A MobileNetV2 encoder was combined with dual split attention and residual classification modules, resulting in approximately 7.3 million parameters. Accuracy, precision, recall, and F1-score of 98.34%, 98.74%, 99.19%, and 99.02%, respectively, were reported. Attention was incorporated architecturally, but a dedicated post-hoc XAI evaluation was not a principal component of the reported analysis. The work successfully addressed unified lightweight multi-crop classification, although relationships among diseases occurring on different crops were not represented explicitly.

A single transfer-learning model for rice, wheat, and bean diseases was developed by Hasan et al. [10]. A combined dataset of 7,037 annotated leaf images from Bangladeshi crop-disease sources and a bean-disease dataset was resized to 224×224, normalized, and augmented using random rotation, horizontal flipping, and zooming. MobileNetV2 was fine-tuned under a 70/15/15 training, validation, and test split and achieved a test accuracy of 97.87%. A dedicated XAI analysis was not reported. The need for three independently trained crop-specific models was reduced by the unified classifier, but sunlight, image quality, environmental variation, and increasing annotation requirements remained limitations, and cross-crop knowledge sharing was not explicitly modelled.

A considerably larger multi-crop benchmark was presented by Zidan et al. [11]. Approximately 45,000 original RGB images from multiple public repositories were curated into 51 healthy and diseased classes covering 12 crops; duplicate and poor-quality samples were manually screened, stratified splitting was used, and controlled training-only augmentation was applied. Eight compact pretrained transformers were benchmarked, followed by the custom PlantaNetLite and PlantaNet CNNs containing approximately 1.28 and 2.58 million parameters. PlantaNet achieved 99.37% validation accuracy and 99.66% test accuracy. Grad-CAM and Grad-CAM++ were used to inspect disease-relevant regions. Strong multi-crop accuracy, explainability, and deployment efficiency were demonstrated, but external field validation remains necessary and the 51 crop–disease classes were learned independently without explicit semantic relations.

E. Graph Neural Networks and Knowledge-Graph Approaches

Graph-based visual aggregation was introduced by Bera et al. [12] through PND-Net. Banana and coffee nutrition-deficiency datasets together with potato and PlantDoc disease datasets were evaluated. Regional features extracted by CNN backbones were summarized through spatial-pyramid pooling and represented as graph nodes for GCN-based feature aggregation. Random rotation, scaling, Gaussian blur, and random cropping were used for augmentation. Accuracies of 96.18% on potato disease and 84.30% on PlantDoc were reported, while Grad-CAM and t-SNE were used for explanation and feature visualization. Graph learning improved regional feature integration, but the graph represented regions derived from images rather than typed entities such as individual leaves, crops, diseases, and disease categories.

Agricultural knowledge and neural reasoning were integrated by Zhao et al. [22] for disease and pest detection in *Elaeagnus angustifolia*. Large language models, an agricultural knowledge graph, representation learning, GNN components, neural-symbolic reasoning, and graph attention were combined using domain-specific disease and pest information. The graph-attention configuration achieved precision of 0.94, recall of 0.92, and accuracy of 0.93. Interpretability was supported by the knowledge structure and neural-symbolic

reasoning, although image-level Grad-CAM was not used as the principal explanation method. The work demonstrated the usefulness of structured agricultural knowledge, but evaluation was restricted to one plant species and the architecture was not designed as a multi-crop leaf-image heterogeneous classifier.

A large agricultural knowledge resource was constructed by Yan et al. [14]. Text and associated information from the Chinese crop diseases and pests image–text database were processed through NLP and entity extraction to construct CropDP-KG. Eight primary entity types and seven relation types were represented, producing 13,840 entities and 21,961 relationships. The resulting graph supported knowledge querying, question answering, overview, and knowledge management rather than direct leaf-image classification. Semantic relations naturally provide human-readable knowledge paths, but no classification-focused post-hoc XAI evaluation was required. The work solved agricultural knowledge organization and retrieval, while image embeddings were not treated as graph nodes and no GNN-based multi-crop image classification was performed.

Visual and semantic information were fused by Alwan and Alturfi [15] using EfficientNet-B8, a Vision Transformer, and Knowledge Graph Fusion. Approximately 87,000 RGB images covering 38 crop–disease classes were used. Crop types, environmental factors, and disease relationships were represented in a knowledge graph, while Node2Vec-style graph embeddings were concatenated with CNN and transformer visual features. Test accuracy of 99.3%, precision of 98.9%, recall of 98.92%, and F1-score of 98.67% were reported. Grad-CAM was used to inspect image-based decisions, while knowledge features provided additional contextual interpretation. The work is closely related to relational plant-disease recognition, but individual leaf images were not represented as typed graph nodes and semantic knowledge was fused as graph embeddings rather than learned through leaf-image HGT message passing.

Leaf-to-leaf relational reasoning was explicitly investigated by Surekha and Subha Mastan Rao [13] through AdjLeafGNN. A curated 10,000-image subset of PlantVillage covering tomato, potato, grape, and apple leaves was normalized, resized to 224×224, and augmented using horizontal and vertical flipping, $\pm 15^{\circ}$ rotation, and brightness variation. CNN embeddings were treated as graph nodes, and K-nearest-neighbour connections with $K = 5$ were formed according to feature-space similarity. The combined attention-enhanced CNN and GNN achieved $98.88\% \pm 0.33\%$ classification accuracy and $98.71\% \pm 0.32\%$ F1-score, while adjacent disease-spread prediction reached an AUC-ROC of 0.942. Attention-based localization was examined,although conventional post-hoc Grad-CAM was not the main explanation method. Image-to-image relational modelling was demonstrated, but all leaf nodes remained the same graph type and explicit crop, disease, and disease-category nodes were absent.

F. Comparative Summary of Related Work

The reviewed literature shows complementary but largely separated research directions. CNN, transformer, and self-supervised methods provide strong visual representations but usually classify each image independently. Multi-crop classifiers place several crops in a common classification space but do not explicitly encode the relationships among crops, diseases, and disease groups. Image-based GNNs mainly model regions or leaf-to-leaf similarity, whereas agricultural knowledge graphs encode semantic entities primarily for retrieval, reasoning, or feature-level fusion. Even the closely related knowledge-graph fusion approach combines visual features with graph embeddings rather than representing each observed leaf image together with crop, disease, and disease-category entities inside a relation-aware HGT. The identified literature therefore does not show the same combination of typed leaf-image, crop, disease, and disease-category nodes, relation-specific HGT message passing, a message-passing/supervision partition designed to reduce direct label shortcutting, controlled zero, one, and two layer message-passing evaluation, and direct comparison between cross-crop and crop-specific graph settings. AgriHGT addresses this gap through a duplicate-aware, support-controlled, transductive multi-crop relational classification framework without assuming that cross-crop relational information must benefit every crop.

III. Method and Materials

A. Overview of the Methodology

AgriHGT was implemented as a two-stage visual–relational classification framework. Fixed features extracted from the adapted visual backbone were used as the initial representations of the `leaf_image` nodes, while learnable representations were assigned to the `crop`, `disease`, and `disease_category` nodes. Heterogeneous message passing was then performed using the Heterogeneous Graph Transformer (HGT) [16]. The complete experimental pipeline is illustrated in Fig. 1

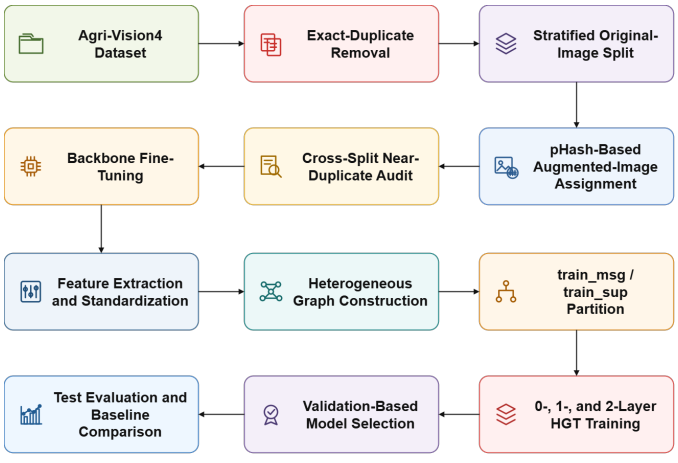


Fig. 1. Overview of the AgriHGT methodology.

B. Dataset

The Agri-Vision Bangladesh dataset [23] was used in this study. It contains leaf images of four crops: Tomato, Papaya, Zucchini, and Bottle Gourd, covering 28 healthy and diseased classes. The images were collected from agricultural fields

in Bangladesh under natural lighting conditions and were validated by agricultural experts. The published dataset contains both original images and augmented images generated using transformations such as rotation, flipping, brightness and contrast adjustment, and noise addition.

The distribution of original and augmented images across the four crops is shown in Fig. 2. A large proportion of the dataset consists of augmented images, which increases the number of training samples available for the different crops.

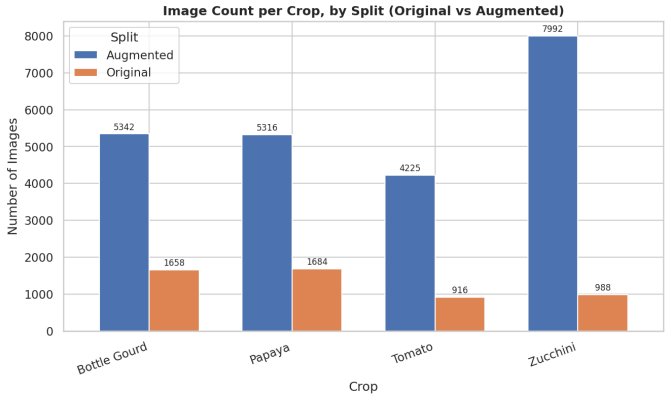


Fig. 2. Distribution of original and augmented images across the four crops in the Agri-Vision dataset.

The dataset contains 28 crop-specific healthy and disease classes. However, the number of images is not equal across all classes. Table. I shows the complete class distribution and highlights this class imbalance. This imbalance was considered during model training and evaluation, and macro-F1 was therefore reported together with classification accuracy.

TABLE I
DISTRIBUTION OF ORIGINAL IMAGES ACROSS THE 28 CROP-SPECIFIC CLASSES

Crop	Class	Images
Bottle Gourd	Mosaic Virus	315
	Alternaria Leaf Blight	303
	Downy Mildew	286
	Anthracnose	276
	Healthy	260
	Early Alternaria Leaf Blight	179
	Fungal Damage Leaf	39
	Total	**1,658**
Papaya	Curled Yellow Spot	538
	Carica Insect Hole	318
	Pathogen Symptoms	286
	Healthy Leaf	189
	Bacterial Blight	183
	Mosaic Virus	119
	Yellow Necrotic Spots/Holes	51
	Total	**1,684**
Tomato	Spot	311
	Healthy	288
	Mosaic	195
	White Spot	65
	Downy	57
	Total	**916**
Zucchini	Yellow Mosaic Virus	202
	Downy Mildew	153
	Anthracnose	129
	Angular Leaf Spot	120
	Healthy	88
	Xanthomonas Leaf Spot	86
	Insect Damage	78
	Dry Leaf	67
	Iron Chlorosis Damage	65
	Total	**988**
	Total Original Images	**5,246**

C. Exact-Duplicate Removal

Exact duplicate removal was performed before the training, validation and test split was constructed. An MD5 hash was calculated for each of the 28,121 image files. Files with identical hashes were grouped, and one representative was retained from each group.

A total of 1,255 byte-identical duplicates were removed. The resulting dataset contained 26,866 images, consisting of 4,304 Original images and 22,562 Augmented images.

D. Duplicate-Aware Initial Split

A random split of the combined Original and Augmented image pools could place an Original image and one of its transformed derivatives in different partitions. Such overlap could result in optimistic validation and test performance. The Original images were therefore used as split anchors.

Class-stratified sampling with a 70/15/15 ratio was first applied to the original images. The initial assignment contained 3,004 training images, 650 validation images, and 650 test images.

Perceptual hashes were subsequently calculated using

$$\texttt{hash_size} = 16,$$

which produced 256-bit hashes. Each Augmented image was compared with Original images belonging to the same crop and class. When the minimum Hamming distance was no greater than six, the Augmented image was linked to the nearest Original image and inherited its partition.

Under this criterion, 1,552 Augmented images were linked to original images. The remaining 21,010 Augmented images that could not be linked confidently to a particular original image and were initially assigned to the training partition.

E. Post-Hoc Near-Duplicate Audit

A same-class post-hoc audit was performed after the initial split had been constructed. Perceptual hashes were compared across different partitions for the following image-pair types:

- Original–Original;
- Original–Augmented; and
- Augmented–Augmented.

When a same-class pair from different partitions had a perceptual-hash Hamming distance of six or less, the validation- or test-side image was reassigned to the training partition. The audit was repeated until no additional cross-partition violation was detected. Successive audit rounds identified 41, 11, 3, and finally 0 violating pairs.

F. Per-Class Support Control

The duplicate audit could remove the only validation or test image of a minority class. A support-control procedure was therefore applied after the audit.

When a class contained no validation image or no test image, an eligible training image from the same class was reassigned to the missing partition. The support-control procedure and same-class near-duplicate audit were then repeated in a bounded loop. A reassigned image could therefore be returned to training when a new cross-partition near-duplicate relation was detected. The final support-controlled split contained 25,122 training images, 896 validation images, and 848 test images.

At least one validation image and one test image were retained for every class. The crop-level distribution was as follows:

- Bottle Gourd: 6,210 training, 300 validation, and 284 test images;
- Papaya: 6,201 training, 281 validation, and 281 test images;
- Tomato: 4,630 training, 186 validation, and 170 test images; and
- Zucchini: 8,081 training, 129 validation, and 113 test images.

The final original-image counts were 3,035 training images, 634 validation images, and 635 test images.

The same final split was used for every backbone experiment. Therefore, differences among the backbone results were not caused by different training, validation, or test partitions.

The partition is described as *duplicate-aware* or *leakage-controlled*, rather than completely leakage-free. The audit was restricted to same-class comparisons and to similarities detected by the selected perceptual-hash representation and threshold. The supported claim is therefore that no same-class cross-partition near-duplicate pair was detected at a pHash Hamming distance of six or less.

G. Visual Backbone Selection

The visual backbone is an important part of AgriHGT because the feature extracted from each leaf image is used as the initial representation of the corresponding `leaf_image` node. The same feature space is also used to construct the `similar_to` relations between leaf images. Therefore, the graph can change when a different backbone is used.

Seven visual backbones were selected to cover different convolutional, transformer, supervised, and self-supervised representation-learning approaches used in recent plant-disease classification studies [6], [9], [10].

- ResNet50;
- DenseNet121;
- EfficientNet-B3;
- ConvNeXt-Tiny;
- Swin-Tiny;
- DINOv2 ViT-B/14; and
- DINOv3 ViT-B/16.

The purpose of using several backbone families was not only to identify the backbone with the highest classification performance. The main aim was also to examine whether heterogeneous graph learning behaves similarly when the initial image representation is changed.

1) ResNet50: ResNet50 was selected as a standard convolutional baseline. Residual connections allow deep convolutional layers to be trained while reducing the difficulty caused by vanishing gradients. ResNet-based transfer learning has also been used for plant-disease recognition [2], [3].

2) DenseNet121: DenseNet121 was selected to represent a different form of convolutional feature learning. Instead of residual addition, DenseNet connects each layer with the outputs of earlier layers. This encourages feature reuse and allows low-level and high-level visual information to remain available throughout the network.This behaviour is useful for leaf-disease images because both small visual details, such as spots and lesions, and larger texture patterns may be important.

3) EfficientNet-B3: This backbone was included because plant-disease symptoms can appear at different spatial scales, while EfficientNet architectures have shown strong performance in leaf-disease classification [4].

4) ConvNeXt-Tiny: ConvNeXt-Tiny was selected as a modern convolutional architecture. Although it remains a CNN, its design includes several ideas influenced by the development of vision transformers. It therefore provides an important middle point between conventional CNNs and transformer-based models. Its inclusion helps determine whether improvements associated with newer visual representations require a transformer architecture or can also be obtained from a modernized convolutional network.

5) Swin-Tiny: Swin-Tiny was selected as a supervised hierarchical vision transformer. Unlike a standard ViT that applies global attention directly across all patches, Swin Transformer performs attention within local windows and gradually combines information through shifted windows and hierarchical stages. Transformer-based and Swin-based models have also been applied to plant-disease recognition [24]–[26].

6) DINOv2 ViT-B/14: DINOv2 ViT-B/14 was selected to represent self-supervised visual pretraining. Unlike the supervised backbones, its original representation learning does not depend on manually assigned image-class labels. Instead, general visual representations are learned from large collections of images using a self-supervised training objective.This is important for AgriHGT because the graph does not require only a strong classifier. It requires an embedding space in which visually and semantically related leaf images can be represented meaningfully. Self-supervised representations may preserve visual structure that is not restricted to the classes used during supervised pretraining. DINOv2 therefore provides a useful comparison between supervised and self-supervised feature spaces.DINOv2 therefore provides a useful comparison between supervised and self-supervised feature spaces, following the growing use of self-supervised learning in plant-disease recognition [7], [8].

7) DINOv3 ViT-B/16: DINOv3 ViT-B/16 was included as a more recent self-supervised vision transformer from the DINO family. It was selected to examine whether a newer self-supervised representation provides stronger leaf-image features for downstream heterogeneous graph learning. The

comparison between DINOv2 and DINOv3 is especially useful because both belong to the self-supervised transformer family. Differences between them therefore provide information about the effect of the quality and generation of the pretrained representation, rather than only comparing a CNN with a transformer.

8) Backbone Categories: The selected backbones can be summarized according to architecture and pretraining strategy as shown in Table II.

H. Backbone Adaptation Before Feature Extraction

The pretrained backbones were not used directly as fixed generic feature extractors. Each backbone was first adapted to the 28-class leaf-disease classification task using only the training partition. This step was performed because features learned from general image datasets may emphasize object shape, background, and general appearance rather than the small colour and texture differences required for disease classification.

A temporary 28-class classification head was attached to each backbone. Most pretrained layers were kept frozen, while only the final representation stage and classification head were fine-tuned. This allowed the high-level representation to become more disease-specific while preserving the general low- and mid-level visual features learned during pretraining.

For ResNet50, `layer4` and the classification head were fine-tuned. For DenseNet121, `denseblock4`, `norm5`, and the classifier were fine-tuned. For EfficientNet-B3, `features[7:]` and the classification head were trained. For ConvNeXt-Tiny, `features[7]` and the classification head were trained. For Swin-Tiny, the final transformer stage, terminal normalization layer, and classification head were fine-tuned. For DINOv2 and DINOv3, the final two transformer blocks, terminal normalization components, and temporary classification head were fine-tuned.

Partial fine-tuning was preferred to complete end-to-end retraining because the dataset is considerably smaller than the datasets used for backbone pretraining. Freezing most of the encoder reduces computational cost and limits unnecessary modification of useful pretrained features.

I. Feature Extraction

During backbone adaptation, validation accuracy was monitored after each epoch. Early stopping was applied to the backbones trained with a patience-based schedule, and training was stopped when validation accuracy did not improve for the predefined number of epochs. The checkpoint corresponding to the highest validation accuracy was saved during training and restored after training was completed. After the best checkpoint was restored, the temporary 28-class classification head was removed or replaced with an identity mapping. The backbone parameters were then frozen, and the adapted models were used only as feature extractors.

A deterministic feature-extraction pass was performed over all 26,866 retained images. Random training augmentation was not applied during this stage. Each image was processed using the input resolution and normalization required by the corresponding backbone.

The resulting feature matrices were:

- ResNet50: $26{,}866 \times 2{,}048$;
- DenseNet121: $26{,}866 \times 1{,}024$;
- EfficientNet-B3: $26{,}866 \times 1{,}536$;
- ConvNeXt-Tiny: $26{,}866 \times 768$;
- Swin-Tiny: $26{,}866 \times 768$;
- DINOv2 ViT-B/14: $26{,}866 \times 768$; and
- DINOv3 ViT-B/16: $26{,}866 \times 768$.

For each backbone, one feature vector was extracted from every retained leaf image. Therefore, the number of rows in the feature matrix was equal to the number of images, while the number of columns represented the embedding dimension of the corresponding backbone.

J. Feature Standardization

The feature values produced by different embedding dimensions can have different ranges and variances. Standardization was therefore performed before graph construction. For each backbone, the mean and standard deviation of every feature dimension were calculated using only the training embeddings. For a feature value x, standardization was performed as

$$x' = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}},$$

where μ_{train} and σ_{train} represent the mean and standard deviation calculated from the training partition. The same training-derived statistics were then applied unchanged to the validation and test features. Therefore, validation and test data were not used to estimate the feature-scaling parameters.

Feature standardization was performed separately for every backbone. A separate heterogeneous graph was then constructed from each standardized feature matrix because the `similar_to` relations depend on distances between backbone embeddings.

The important point is that the seven backbones do not only provide seven different inputs to the same fixed graph. Each backbone produces a different visual feature space, and that feature space determines both the initial `leaf_image` representation and the nearest-neighbour structure used by the graph. The backbone comparison therefore examines how different visual representations interact with heterogeneous message passing.

K. Heterogeneous Graph Construction

After feature extraction and standardization, a separate heterogeneous graph was constructed for each backbone representation, following the growing use of graph and knowledge-based modelling in plant-disease analysis [12], [22], [27].. The graph was designed to combine visual information from individual leaf images with crop identity, disease identity, and broader disease relationships.

Four node types were defined:

- `leaf_image`: one node for each retained image;
- `crop`: one node for each crop species;

TABLE II
VISUAL BACKBONE CATEGORIES AND THEIR ROLE IN AGRIHGT

Backbone	Architecture	Pretraining Type	Reason for Inclusion
ResNet50	Residual CNN	Supervised	Standard CNN reference
DenseNet121	Densely connected CNN	Supervised	Feature reuse and alternative CNN representation
EfficientNet-B3	Compound-scaled CNN	Supervised	Efficient depth, width, and resolution scaling
ConvNeXt-Tiny	Transformer-inspired modern CNN	Supervised	Bridge between conventional CNNs and modern designs
Swin-Tiny	Hierarchical vision transformer	Supervised	Local-window and hierarchical transformer representation
DINOv2 ViT-B/14	Vision Transformer	Self-supervised	Strong self-supervised visual representation
DINOv3 ViT-B/16	Vision Transformer	Self-supervised	Recent self-supervised representation from the DINO family

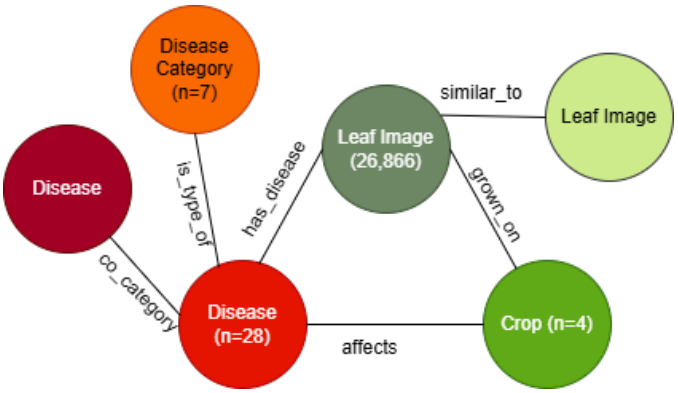


Fig. 3. Basic semantic structure of the graph.

- `disease`: one node for each crop-specific disease or healthy class; and
- `disease_category`: one node for each broader disease group.

The resulting graph contained 26,866 `leaf_image` nodes, four `crop` nodes, 28 `disease` nodes, and seven `disease_category` nodes, giving a total of 26,905 nodes. The seven disease categories were Healthy, Viral, Bacterial, Fungal, Pest, Nutrient Deficiency, and Other. Each of the 28 crop-specific classes was assigned to one of these broader categories. This additional level was introduced so that diseases belonging to different crops could still share a higher-level semantic relation when they represented a similar type of disorder, consistent with recent agricultural knowledge-graph modelling [14], [28]. The standardized backbone feature of each image was assigned as the initial feature of the corresponding `leaf_image` node. The non-image nodes do not have natural visual features; therefore, learnable node-specific representations were used for crop, disease, and disease-category nodes during HGT training. Fig 3 shows the basic semantic structure of the graph.

1) Graph Relations: Six forward relation types were defined:

1) `grown_on`: connects each leaf image to its crop;
2) `affects`: connects each crop to its associated disease classes;
3) `is_type_of`: connects each disease to its broader disease category;
4) `similar_to`: connects visually similar leaf images;
5) `has_disease`: connects selected training images to their corresponding disease nodes; and
6) `co_category`: connects disease nodes belonging to the same disease category.

2) Visual Similarity Relation: The `similar_to` relation was constructed from the standardized backbone embeddings. For every image, the five nearest neighbours were identified using cosine distance,

$$d_{\cos}(\mathbf{x}_i, \mathbf{x}_j) = 1 - \frac{\mathbf{x}_i^T \mathbf{x}_j}{\|\mathbf{x}_i\|_2 \|\mathbf{x}_j\|_2}.$$

Neighbour search was restricted to images belonging to the same crop. This restriction was used because differences in leaf shape and overall appearance between crop species can dominate feature-space similarity. For every `leaf_image` node, five directed textttsimilar_to edges were created. With 26,866 leaf images, this produced 134,330 forward similarity edges. The neighbours were generated independently for every backbone because each backbone defines a different visual feature space.

L. Message-Passing and Supervision Partition

A direct `has_disease` edge cannot safely be assigned to every training image. If an image contributing to the classification loss were also directly connected to its true disease node, reverse graph relations could create the shortcut

$$\texttt{leaf_image} \rightarrow \texttt{disease} \rightarrow \texttt{leaf_image},$$

allowing the correct class information to return to the same image during message passing.

The training partition was therefore divided into two disjoint roles. The partition was created independently within each disease class using a 75/25 message-passing-to-supervision ratio. Original images and their linked Augmented derivatives were treated as a single group so that near-duplicate members remained on the same side of the partition.

The resulting fixed partition contained 18,881 `train_msg` and 6,241 `train_sup` images.

The two subsets were used differently:

- `train_msg` images retained their `has_disease` edges. Their visual representations were allowed to contribute to disease-node representations through message passing, but their predictions were excluded from the classification loss.

- `train_sup` images had no `has_disease` edges. Only these training images contributed to the supervised classification loss.

Validation and test images were also not assigned `has_disease` edges. Consequently, `train_sup`, validation, and test images followed the same rule regarding access to their true disease relation. This design removes the explicit disease-label shortcut from supervised examples and still allows disease nodes to receive visual grounding from the `train_msg` subset.

Every declared graph relation was initially created as a directed edge. The pyHGT graph implementation additionally generated a separate reverse relation. For example,

$$\texttt{leaf_image} \xrightarrow{\texttt{grown_on}} \texttt{crop}$$

was accompanied by

$$\texttt{crop} \xrightarrow{\texttt{rev_grown_on}} \texttt{leaf_image}.$$

Separate relation identifiers were retained for the forward and reverse directions, allowing HGT to learn different transformations for the two directions. A self-loop was also added to every node before tensor conversion. The final graph therefore contained 13 relation types, including forward relations, reverse relations, and the self relation.

The heterogeneous graph was converted to PyTorch tensors using the pyHGT graph representation. Node features, node-type identifiers, edge indices, edge-type identifiers, and edge-time values were produced for full-batch processing. No temporal information was available in the dataset. Therefore, all edge-time values were fixed and HGT temporal encoding was disabled. The complete graph contained 26,905 nodes. After reverse relations and self-loops were included, 387,301 graph edges were supplied to the HGT model.

Crop, disease, and disease-category nodes do not contain pixel-derived features comparable to the backbone embeddings of the leaf images. Assigning the same fixed zero vector to every node of the same type would make structurally similar semantic nodes difficult to distinguish. So, a separate learnable embedding was therefore assigned to every non-image node. Let m denote a crop, disease, or disease-category node. Its input representation was

$$\mathbf{x}_m = E_m,$$

where E_m is a trainable embedding vector. These parameters were updated together with the HGT model. The `leaf_image` nodes were not given learnable identity embeddings as their standardized backbone features were retained as their visual input.

M. AgriHGT Model Architecture

After graph construction, the complete heterogeneous graph was passed to the AgriHGT model. The model contains three main parts: HGT message passing, gated fusion of visual and graph features, and a leaf–disease decoder. The complete model structure is shown in Fig. 4.

1) Node Projection and HGT Message Passing: The input features of different node types do not have the same form. Leaf-image nodes contain standardized backbone features, while crop, disease, and disease-category nodes use learnable embeddings. Before message passing, all node representations were mapped to the same 256-dimensional space.

For a node v of type $\tau(v)$, the initial hidden representation was

$$\mathbf{h}_v^{(0)} = \tanh\left(W_{\tau(v)}\mathbf{x}_v + \mathbf{b}_{\tau(v)}\right),$$

where $\mathbf{x}_v$ is the input feature and $W_{\tau(v)}$ is the transformation for that node type.

HGT was used because the graph contains several types of nodes and relations [16]. A message coming through a `similar_to` edge should not be treated in the same way as a message coming through a `grown_on` or `is_type_of` edge. A message coming through a `similar_to` edge should not be treated in the same way as a message coming through a `grown_on` or `is_type_of` edge. HGT therefore uses node-type and relation-type information when calculating attention. For a source node u, target node v, and relation r, an attention score can be written as

$$e_{uv} = \frac{\mathbf{q}_v^T W_r \mathbf{k}_u}{\sqrt{d}},$$

where $\mathbf{q}_v$ and $\mathbf{k}_u$ are the query and key representations, W_r is a relation-specific transformation, and d is the attention dimension.

The attention weights were obtained using

$$\alpha_{uv} = \frac{\exp(e_{uv})}{\sum_{k\in\mathcal{N}(v)}\exp(e_{kv})}.$$

Neighbour information was then combined according to these weights. Four attention heads were used. The hidden dimension was 256 and the HGT dropout rate was 0.2.

Three message-passing depths were tested:

$$n_{\text{layers}} \in \{0, 1, 2\}.$$

The zero-layer model skipped HGT message passing. The node projections, learnable non-image embeddings, gated fusion, and disease decoder were still kept. The zero-layer model was therefore used to examine model performance without graph propagation.

2) Gated Residual Fusion: The backbone already provides useful visual information about each leaf. This information may be changed during graph message passing. A direct visual path was therefore kept and combined with the HGT output.

The original standardized image feature was first projected to the same 256-dimensional space:

$$\mathbf{r}_i = \text{LayerNorm}\left(W_r\mathbf{x}_i + \mathbf{b}_r\right).$$

Let $\mathbf{h}_i$ be the HGT output for leaf image i. A learnable gate was calculated as

$$\mathbf{g}_i = \sigma\left(W_g[\mathbf{r}_i \| \mathbf{h}_i] + \mathbf{b}_g\right),$$

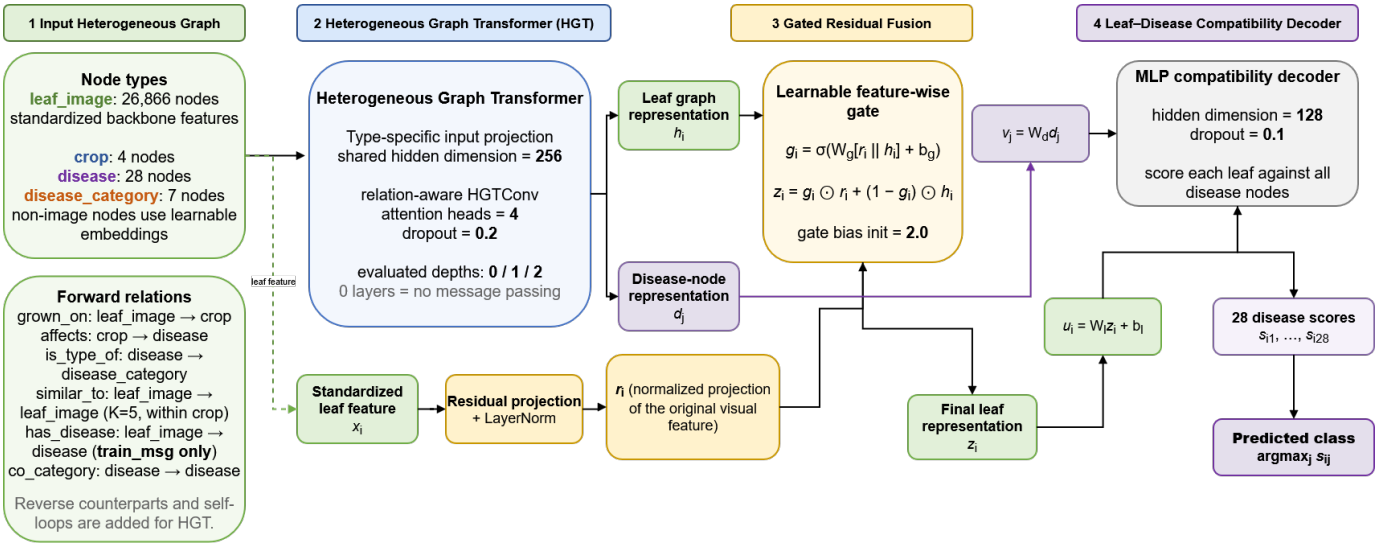


Fig. 4. Architecture of AgriHGT.

where $\|$ denotes concatenation and σ is the sigmoid function.

The final leaf representation was

$$\mathbf{z}_i = \mathbf{g}_i \odot \mathbf{r}_i + (1 - \mathbf{g}_i) \odot \mathbf{h}_i.$$

The gate bias was initialized to 2.0. Since

$$\sigma(2) \approx 0.88,$$

the model initially placed more weight on the original visual feature. The amount of graph information used was then learned during training.

3) Leaf–Disease Compatibility Decoder: The final leaf representation was compared with all 28 disease-node representations. A small multilayer perceptron was used for this step.

The leaf and disease representations were first projected as

$$\mathbf{u}_i = W_l \mathbf{z}_i + \mathbf{b}_l,$$

and

$$\mathbf{v}_j = W_d \mathbf{d}_j,$$

where $\mathbf{d}_j$ is the representation of disease node j.

The score between leaf i and disease j was calculated as

$$s_{ij} = W_o \, \mathrm{Dropout}\left[\mathrm{ReLU}\left(\mathbf{u}_i + \mathbf{v}_j\right)\right] + b_o.$$

A hidden dimension of 128 and a dropout rate of 0.1 were used in the decoder. Since 28 disease nodes were present, 28 scores were produced for every leaf image. The predicted disease was

$$\hat{y}_i = \arg\max_j s_{ij}.$$

N. Model Training and Optimization

AgriHGT was trained using the complete graph in full-batch mode. The features and graph connections of training, validation, and test nodes were present during each forward pass. However, the classification loss was calculated only from the `train_sup` images. `train_msg`, validation, and test labels were not used in the loss.

Weighted cross-entropy was used because the number of images differs across the 28 classes. Class weights were calculated only from the `train_sup` labels. For class c, an inverse-frequency weight was obtained from

$$\tilde{w}_c = \frac{1}{n_c},$$

where n_c is the number of `train_sup` images in class c. The weights were then normalized to have a mean of one.

The training loss was

$$\mathcal{L} = -\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} w_{y_i} \log \left(\frac{\exp(s_{i,y_i})}{\sum_{j=1}^{28} \exp(s_{ij})} \right),$$

where $\mathcal{S}$ is the `train_sup` set and y_i is the true class.

AdamW was used with a learning rate of 1×10^{-3} and weight decay of 1×10^{-4}. Gradient norms were clipped at 1.0. A cosine learning-rate schedule reduced the learning rate to 1×10^{-6}.

The training settings are summarized in Table III.

For each backbone, zero, one, and two layer HGT models were trained under the same data split. The checkpoint with the highest validation accuracy was saved. A minimum improvement of 10^{-4} was required. The HGT depth was also selected using validation accuracy. The test set was used only after the model and HGT depth had been selected.

The maximum HGT training epoch was set to 300 epochs for ResNet50, Swin-Tiny, and DenseNet121, and 400 epochs for ConvNeXt-Tiny, EfficientNet-B3, DINOv2, and DINOv3. Validation accuracy was monitored throughout training, and

TABLE III
AGRIHGT TRAINING SETTINGS

Parameter	Value
Hidden dimension	256
Attention heads	4
HGT layers	0, 1, 2
HGT dropout	0.2
Decoder hidden dimension	128
Decoder dropout	0.1
Gate bias	2.0
Nearest neighbours (K)	5
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Gradient clipping	1.0
Minimum learning rate	1×10^{-6}
Main random seed	42

the checkpoint with the highest validation accuracy was retained. Early stopping with a patience of 50 epochs was used for the 400-epoch configurations, whereas the 300-epoch configurations used a patience equal to the full training budget.

1) Transductive Training Setting: The experiments followed a transductive setting. Validation and test images were present as graph nodes during HGT training, so their image features and graph connections could take part in message passing. Their true disease labels were not used for optimization, and no `has_disease` edge was provided for these images.

The reported results therefore measure transductive node classification. They do not represent a fully inductive setting in which completely new graph nodes are introduced only after training.

O. Reference Model and Ablation Design

Two types of reference experiments were used: a feature-only logistic regression model and HGT ablation experiments.

1) Feature-Only Reference Model: A multinomial logistic-regression classifier was trained separately for each visual backbone. The same standardized backbone features were used, but no graph was provided.

The logistic-regression model therefore did not use crop nodes, disease nodes, disease-category nodes, graph relations, HGT message passing, gated fusion, or the leaf–disease decoder. All 25,122 training images were used because no graph-based label shortcut was present. A maximum of 2,000 optimization iterations was used.

This model provides a reference for the performance that can be obtained from the visual features alone, as commonly used in conventional image-based plant-disease classifiers [1], [29].

The zero-layer HGT model was used as a second reference. Unlike logistic regression, it retained the node projections, non-image node embeddings, gated fusion, and disease decoder. Only HGT message passing was removed. For this reason, comparison between zero-layer HGT and one- or two-layer HGT gives a more direct view of the effect of message passing.

2) Graph-Component Ablation: A controlled ablation study was performed using DINOv3 ViT-B/16 features to examine the contribution of the main structural and architectural components of AgriHGT. The two-layer HGT configuration selected according to validation accuracy in the main DINOv3 experiment was retained for the ablation study, except for the explicit no-HGT configuration.

Nine configurations were evaluated:

1) **A0 - Full AgriHGT:** all graph and model components were retained;
2) **A1 - No HGT message passing:** HGT message passing was removed by setting the number of HGT layers to zero;
3) **A2 - No leaf-similarity edges:** `similar_to` and reverse-`similar_to` relations were removed;
4) **A3 - No crop knowledge:** crop nodes together with `grown_on`, reverse-`grown_on`, `affects`, and reverse-`affects` relations were removed;
5) **A4 - No disease taxonomy:** disease-category nodes together with `is_type_of`, reverse-`is_type_of`, `co_category`, and reverse-`co_category` relations were removed;
6) **A5 - No disease visual grounding:** `has_disease` and reverse-`has_disease` relations were removed from the `train_msg` images;
7) **A6 - No learnable meta-node identity:** the learnable crop, disease, and disease-category node embeddings were replaced with fixed zero vectors;
8) **A7 - No residual feature bypass:** the direct projected DINOv3 visual representation was removed and only the HGT-updated leaf representation was used; and
9) **A8 - No learned fusion gate:** the learnable feature-wise gate was replaced with an equal-weight average of the visual and graph representations.

The dataset split, DINOv3 feature bank, and training configuration were kept fixed so that each setting represented a controlled change from the full model. A fixed random seed of 42 was used. The ablation models were trained for a maximum of 400 epochs with an early-stopping patience of 50 epochs (same as before). The hidden dimension was 256, four attention heads were used, and the dropout rate was 0.2. AdamW with a learning rate of 1×10^{-3} and weighted cross-entropy loss was retained.

The no-HGT configuration, A1, was taken from the zero-layer model evaluated during the main HGT-depth experiment rather than being trained again. Test accuracy and macro-F1 were used to compare each ablation configuration with the full AgriHGT model.

P. Evaluation Metrics

Performance was measured using accuracy, precision, recall, and macro-F1. A confusion matrix was also used to examine class-level errors.

For N test images, accuracy was calculated as

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{N}.$$

For class c, precision was calculated as

$$P_c = \frac{TP_c}{TP_c + FP_c},$$

where TP_c is the number of true positives and FP_c is the number of false positives.

Recall was calculated as

$$R_c = \frac{TP_c}{TP_c + FN_c},$$

where FN_c is the number of false negatives.

The F1-score for class c was

$$F1_c = \frac{2P_cR_c}{P_c + R_c}.$$

Since the dataset is imbalanced, macro-F1 was also reported. It gives equal weight to all 28 classes:

$$F1_{\text{macro}} = \frac{1}{28}\sum_{c=1}^{28} F1_c.$$

Per-class precision, recall, F1-score, and support were also calculated. Support is the number of test images belonging to a class.

Q. Experimental Setup

The experiments were run in the Kaggle notebook environment using a single NVIDIA Tesla T4 GPU. PyTorch 2.10.0 with CUDA 12.8 support was used for model training.

PyTorch and Torchvision were used for backbone training and feature extraction. The official pyHGT implementation was used for the heterogeneous graph structure and HGT layers. PyTorch Geometric was used for supporting graph operations. Scikit-learn was used for nearest-neighbour search, evaluation functions, and logistic regression. Pandas and NumPy were used for data processing. Pillow and ImageHash were used for image handling and duplicate checking.

The support-controlled split was fixed for the main experiments and contained 25,122 training, 896 validation, and 848 test images. The main HGT experiments used seed 42. A fixed hold-out validation set was used rather than k-fold cross-validation. Validation accuracy was used to select both the best training checkpoint and the number of HGT layers.

The main backbone experiments were single-seed experiments. The graph-component and cross-crop ablation experiments were repeated using five seeds so that the mean and standard deviation could be reported.

IV. Results and Discussion

A. Experimental Results

1) Performance of Supervised-Pretrained Backbones: The backbones were first grouped according to their pretraining method. ResNet50, ConvNeXt-Tiny, EfficientNet-B3, Swin-Tiny, and DenseNet121 were initialized from supervised pretrained weights. Their validation-selected AgriHGT results and corresponding feature-only logistic-regression references are presented in Table IV.

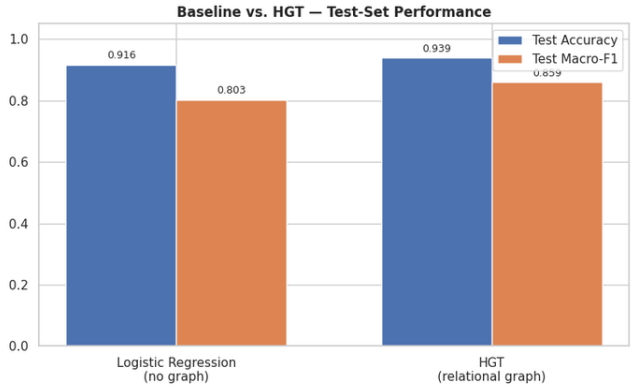


Fig. 5. Logistic Regression vs. AgriHGT comparison.

Among the supervised-pretrained models, the effect of heterogeneous graph modelling was mixed. ResNet50 and DenseNet121 obtained small improvements in test accuracy, while ConvNeXt-Tiny produced almost the same test accuracy as its feature-only reference. EfficientNet-B3 performed slightly better with logistic regression than with its validation-selected AgriHGT configuration. Swin-Tiny also produced the same test accuracy in both settings, although its feature-only reference obtained a higher macro-F1.

2) Performance of Self-Supervised Backbones: DINOv2 ViT-B/14 and DINOv3 ViT-B/16 were evaluated separately because their pretrained representations were learned using self-supervised learning rather than supervised image-class labels. The results are shown in Table V.

Both self-supervised backbones benefited from heterogeneous graph modelling. DINOv2 increased from 0.9175 to 0.9257 in test accuracy and from 0.8069 to 0.8374 in macro-F1. A larger improvement was obtained with DINOv3. Its test accuracy increased from 0.9163 to 0.9387, while macro-F1 increased from 0.8027 to 0.8588 (Fig. 5). The two-layer DINOv3 model therefore produced the strongest observed result in the study.

3) Effect of HGT Message-Passing Depth: Zero, one, and two layer HGT configurations were evaluated. A zero-layer configuration retained the projection, residual fusion, and compatibility decoder but performed no graph message passing. Model depth was selected according to validation accuracy.

For the supervised-pretrained backbones, the most suitable depth varied between models. One layer was selected for ResNet50, EfficientNet-B3, and DenseNet121, while two layers were selected for ConvNeXt-Tiny and Swin-Tiny. The test-optimal depth was not always the validation-selected depth, and the test results were not used for model selection.

A clearer progression was observed for DINOv3:

Test accuracy increased from 0.9175 without message passing to 0.9328 with one HGT layer and 0.9387 with two layers. Macro-F1 also increased from 0.8119 to 0.8345 and then to 0.8588. This result shows that the DINOv3 representation benefited progressively from relational message passing in the main experiment.

4) Training Behaviour: The training behaviour of the selected two-layer DINOv3 model is shown in Fig. 6.

TABLE IV
PERFORMANCE OF SUPERVISED-PRETRAINED VISUAL BACKBONES

Backbone	Model	HGT Layers	Val. Acc.	Test Acc.	Macro-F1
ResNet50	Logistic Regression	–	0.8962	0.8844	**0.7821**
	AgriHGT	1	**0.9118**	**0.8927**	0.7698
ConvNeXt-Tiny	Logistic Regression	–	0.8996	0.8998	0.8064
	AgriHGT	2	**0.9129**	0.8998	**0.8071**
EfficientNet-B3	Logistic Regression	–	0.8862	**0.8998**	**0.7996**
	AgriHGT	1	**0.9029**	0.8915	0.7973
Swin-Tiny	Logistic Regression	–	0.8894	0.9055	**0.8045**
	AgriHGT	2	**0.9128**	0.9055	0.7828
DenseNet121	Logistic Regression	–	0.8514	0.8512	**0.7619**
	AgriHGT	1	**0.8704**	**0.8548**	0.7616

TABLE V
PERFORMANCE OF SELF-SUPERVISED DINO BACKBONES

Backbone	Model	HGT Layers	Val. Acc.	Test Acc.	Macro-F1
DINOv2 ViT-B/14	Logistic Regression	-	0.8951	0.9175	0.8069
	AgriHGT	1	**0.9219**	**0.9257**	**0.8374**
DINOv3 ViT-B/16	Logistic Regression	-	0.9040	0.9163	0.8027
	AgriHGT	2	**0.9263**	**0.9387**	**0.8588**

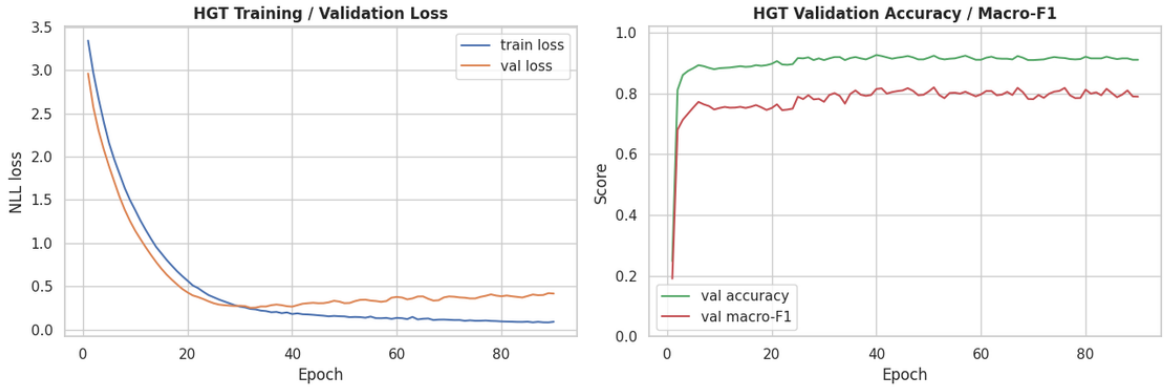


Fig. 6. Training behaviour of the selected two-layer DINOv3 AgriHGT model.

TABLE VI
EFFECT OF HGT DEPTH WITH DINOV3 FEATURES

Layers	Val. Acc.	Test Acc.	Macro-F1
0	0.9107	0.9175	0.8119
1	0.9241	0.9328	0.8345
2*	**0.9263**	**0.9387**	**0.8588**

Training loss decreased rapidly during the early epochs, while validation accuracy increased and reached a maximum of 0.9263. Validation loss later increased as training loss continued to decrease, indicating increasing overfitting. Early stopping occurred at epoch 90, and the checkpoint with the highest validation accuracy was retained for test evaluation.

The selected DINOv3 model performed strongly on most Bottle Gourd, Papaya, and Tomato classes. Several classes obtained F1-scores close to or equal to 1.000. Lower performance was mainly concentrated among Zucchini disease classes.

Zucchini Angular Leaf Spot was one of the most difficult classes. It had 36 test samples but achieved a precision of 0.684, recall of 0.361, and F1-score of 0.473. Zucchini Anthracnose also obtained a low F1-score of 0.381. Since these classes contained more test samples than several other minority classes, their weaker performance cannot be explained by low support alone. Visual similarity among Zucchini symptoms may also contribute to these errors.

Classes containing only one or two test samples, including Zucchini Healthy, Iron Chlorosis Damage, and Xanthomonas Leaf Spot, should be interpreted cautiously because a single prediction can strongly affect their class-wise scores.

The corresponding confusion matrix is shown in Fig. 7.

5) Qualitative and Explainability Analysis: Grad-CAM visualizations were examined to provide a qualitative view of the visual information used by the DINOv3 backbone.

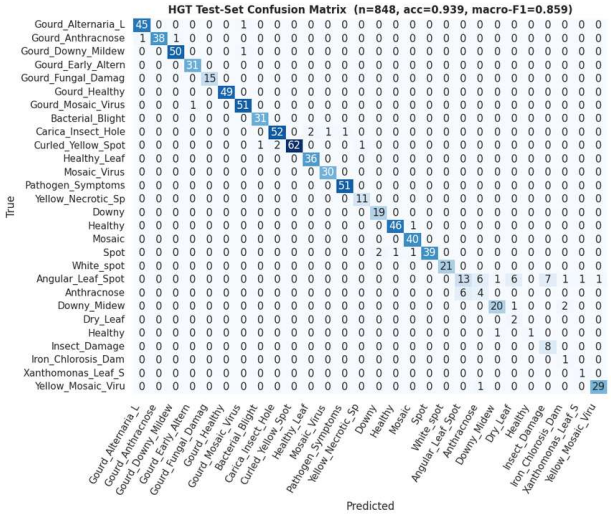


Fig. 7. Test confusion matrix of the selected two-layer DINOv3 AgriHGT model.

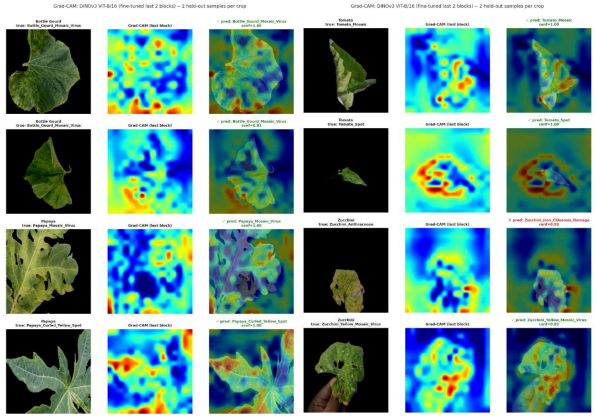


Fig. 8. Representative Grad-CAM visualizations for some selected DINOv3 leaf-disease predictions.

Representative examples are shown in Fig. 8. The activation maps highlight image regions associated with visible disease patterns such as discoloration, spots, lesions, and damaged leaf areas. Selected misclassified samples are also shown in Fig. 9. These examples provide additional insight into difficult cases where visually similar symptoms may lead to incorrect predictions.

6) DINOv3 Component Ablation: A controlled ablation study was conducted using DINOv3 ViT-B/16 features to examine the contribution of the main structural and architectural components of AgriHGT. Nine configurations were evaluated. The full model, A0, was used as the internal reference for this experiment. A1 reused the zero-layer result, while A2–A8 each removed or replaced one component of the two-layer model.

The ablation results are presented in Table VII. This ablation was conducted using one fixed seed and should therefore be interpreted as a controlled component analysis rather than a statistical stability experiment.

The largest decrease in both accuracy and macro-F1 occurred when the residual feature bypass was removed in A7. Test accuracy decreased by 0.94 percentage points and macro-F1 decreased by 1.00 percentage point relative to A0. This indicates that direct preservation of the DINOv3 visual representation was important and that graph propagation alone was not sufficient to retain all useful visual information.

A8 also produced lower accuracy and macro-F1 when the learned gate was replaced with a fixed equal-weight average. This result indicates that adaptive fusion between the original visual representation and the graph-updated representation was more useful than assigning both representations the same fixed weight.

Removal of disease visual grounding in A5 reduced both metrics. Therefore, the `has_disease` relations provided useful information for connecting disease nodes with visual disease evidence.

The effects of several structural relations were mixed. Removal of leaf-similarity edges in A2 reduced accuracy but slightly increased macro-F1. Removing crop knowledge in A3 also reduced accuracy while increasing macro-F1. Therefore, these components did not improve all classes uniformly.

An important result was obtained in A4. Removing disease-category nodes, `is_type_of`, and `co_category` relations increased test accuracy from 0.9328 to 0.9340 and macro-F1 from 0.8219 to 0.8535. Thus, the disease taxonomy did not improve the DINOv3 model in this ablation run. The coarse disease-category structure may have introduced relations that were not always aligned with the fine-grained visual differences required for the 28-class classification task. This result should be treated as an observed single-seed finding rather than a general conclusion.

TABLE VII
DINOv3 ARCHITECTURAL ABLATION OF AGRIHGT

ID	Configuration	Test Acc.	Macro-F1	ΔAcc.	ΔF1
A0	Full AgriHGT	0.9328	0.8219	-	-
A1	No HGT message passing	0.9281	0.8300	-0.0047	+0.0081
A2	No leaf-similarity edges	0.9257	0.8253	-0.0071	+0.0034
A3	No crop knowledge	0.9269	0.8359	-0.0059	+0.0140
A4	No disease taxonomy	**0.9340**	**0.8535**	+0.0012	+0.0316
A5	No disease visual grounding	0.9292	0.8167	-0.0035	-0.0052
A6	No learnable meta-node identity	0.9269	0.8278	-0.0059	+0.0059
A7	No residual feature bypass	0.9233	0.8119	-0.0094	-0.0100
A8	No learned fusion gate	0.9304	0.8175	-0.0024	-0.0044

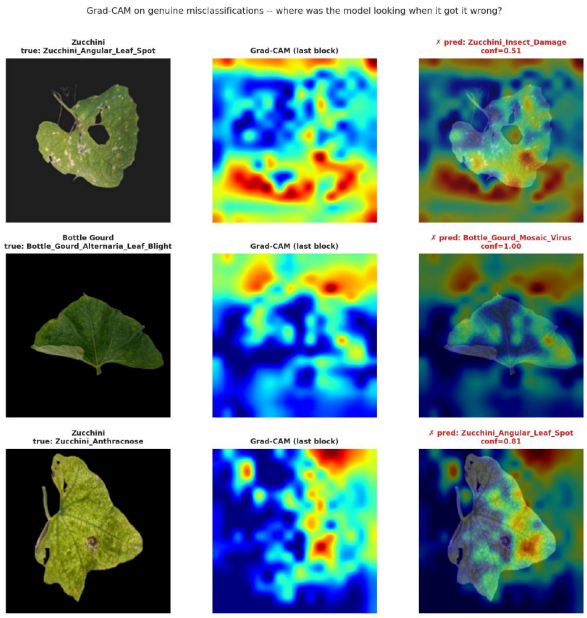


Fig. 9. Representative misclassified leaf images from the DINOv3 model, showing difficult cases with visually similar disease symptoms.

The A0 result in this controlled ablation run differs from the main DINOv3 result because the ablation was executed as a separate controlled experiment. Therefore, the ablation configurations are interpreted relative to A0 within this experiment rather than directly against the main 0.9387 test result.

B. Discussion

1) Best-Performing Model and Technical Interpretation: The strongest result was obtained by the two-layer DINOv3 AgriHGT model, with 93.87% test accuracy and 85.88% macro-F1. This was higher than the DINOv3 feature-only reference, showing that HGT added useful relational information beyond the visual features alone.

The self-supervised DINO backbones also showed more consistent gains from graph modelling than the supervised-pretrained backbones [7], [8]. Detailed visual patterns such as texture, shape, discoloration, and lesion structure are important for separating visually similar leaf diseases [25], [30], [31]. In AgriHGT, these features are important both as `leaf_image` node inputs and for building `similar_to` connections.

The depth results support this finding, as DINOv3 accuracy increased from 91.75% with no HGT layers to 93.87% with two layers. The ablation study also showed that the residual visual path and learned fusion gate were important, while the disease taxonomy did not consistently improve performance. This suggests that the strongest results came from combining high-quality DINOv3 features with selective graph information rather than from adding every graph component.

2) Comparison with Published Studies: Table VIII compares AgriHGT with representative graph-based, transformer-based, and hybrid plant-disease approaches [12], [13], [32], [33]. Direct performance ranking should be avoided because the studies use different datasets, class distributions, image conditions, and evaluation protocols.

The numerical values in Table VIII should not be interpreted as direct evidence that one method is superior to another, as plant-disease studies use different datasets and evaluation settings [17], [29].For example, several higher accuracies were reported on PlantVillage [5], [19], while AgriHGT was evaluated on Agri-Vision4 with four crops and 28 crop-specific classes using a duplicate-aware and support-controlled split. The main distinction of AgriHGT is therefore not a direct accuracy advantage over studies using different datasets, but the use of a heterogeneous graph in which leaf images, crops, diseases, and disease categories are represented as different node types and jointly processed through HGT [14], [22], [28].

3) Error Analysis: The main errors were found in the Zucchini classes. Zucchini Angular Leaf Spot had an F1-score of 0.473, while Zucchini Anthracnose had an F1-score of 0.381.This may be caused by similar visual symptoms across different disease classes, which remains a challenge in fine-grained plant-disease recognition [32], [34].

Some Zucchini classes also had only one or two test samples, so their scores can change greatly with a single prediction. The gap between 93.87% accuracy and 85.88% macro-F1 also shows that minority and difficult classes still affect the overall performance.

TABLE VIII
COMPARISON WITH REPRESENTATIVE PUBLISHED PLANT-DISEASE STUDIES

Study	Dataset	Method	Reported Performance	Limitation / Main Difference
Wang et al. [7]	CCMT + collected dataset	SSL (MAE + CBAM)	95.35 / 99.61% Acc.	Self-supervised visual learning; no graph relations
Zhao et al. [8]	PlantVillage + PlantDoc + augmented data	Contrastive SSL + domain adaptation	90.52% Acc.	Focuses on limited labels and domain shift
Thakur et al. [6]	Five public datasets	CNN + ViT	98.86% Acc. on PlantVillage	Visual hybrid model without structured crop–disease relations
Subburaj et al. [9]	Five-crop datasets	Plantention	98.34% Acc., 99.02% F1	Lightweight multi-crop image classifier without graph modelling
Hasan et al. [10]	Rice, Wheat, and Bean	MobileNetV2 transfer learning	97.87% Acc.	Multi-crop classifier based only on visual features
Alwan and Alturfi [15]	38 plant-disease categories	EfficientNet-B8 + ViT + KGF	99.30% test Acc.	Fuses external knowledge rather than typed image nodes in HGT
AgriHGT	**Agri-Vision4**	**DINOv3 + HGT**	**93.87% Acc., 85.88% Macro-F1**	**Typed leaf-image, crop, disease, and category graph**

4) Robustness and Sensitivity: The HGT-depth experiment showed that DINOv3 improved as message passing increased from zero to two layers. The two-layer model was selected using validation accuracy.

The ablation results also showed that some components were more important than others. Removing the residual bypass, disease grounding, or learned fusion gate reduced performance, while other removals had mixed effects.

Since these experiments used fixed seeds and no statistical significance test was performed, small differences should be treated as observed trends rather than confirmed improvements.

5) Practical Implications and Limitations: AgriHGT enables Tomato, Papaya, Zucchini, and Bottle Gourd diseases to be modeled within a single heterogeneous graph using image, crop, disease, and relation information. DINOv3 features combined with HGT provided strong performance by preserving the original visual representation while also adding graph-based relational information. Cross-crop and cross-dataset generalization remain important challenges [17], [21]. In contrast, the current AgriHGT evaluation is transductive and does not evaluate completely unseen crop domains.

The current setting is transductive, so new field images would need to be added to the graph and connected to suitable neighbours before prediction. Future work should therefore evaluate an inductive setting, increase data for minority Zucchini classes, and further refine the disease taxonomy.

V. CONCLUSION

AgriHGT was developed for multi-crop leaf-disease classification by combining visual features with a heterogeneous graph containing leaf images, crops, diseases, and disease categories. Seven visual backbones were evaluated, and the strongest observed result was obtained with DINOv3 ViT-B/16 and two HGT layers, achieving **93.87% test accuracy** and **85.88% macro-F1**. The results also showed that HGT message passing, the residual visual path, and learned fusion contributed to the final performance. However, the study was limited to four crops and 28 classes, several Zucchini classes had very few test samples, the main experiments used fixed random seeds, and the evaluation was transductive. Future work should therefore test AgriHGT in an inductive setting, include more samples for minority classes, refine the disease taxonomy, perform multi-seed and cross-validation experiments, evaluate the model on independent datasets, and investigate lighter backbones and more efficient graph construction for practical deployment.

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